

QIMING CUI

Applied Mathematics and Statistics Dept., Johns Hopkins University, Maryland, USA 21218

☎ (+1) 667-263-9069 ✉ qcui6@jhu.edu  [LinkedIn](#)

CANDIDATE INFORMATION

- 4th year Ph.D. Candidate in Applied Mathematics and Statistics, Johns Hopkins University
- Research group: Algorithms and Complexity (CS Theory group)
- Research focus: Approximation and Online Algorithms with Machine-Learned Predictions

EDUCATIONAL BACKGROUND

Johns Hopkins University

Doctor of Philosophy in Applied Math and Statistics

August 2027 (expected)

Adviser: Dr. Michael Dinitz

University of Oxford

Master of Science in Mathematical Sciences

July 2022

Graduate with Merit

University of Liverpool

Bachelor of Science with Honours in Mathematics (First Class)

June 2021

GPA: 4.0/4.0

RELEVANT COURSEWORK

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|----------------------------|-----------------------|------------------------------|----------------------|
| • Algorithms | • Probability Theory | • Continuous Optimization | • Real Analysis |
| • Approximation Algorithms | • Statistics Theory | • Combinatorial Optimization | • Numerical analysis |
| • Machine Learning | • Convex Optimization | • Matrix Analysis | • Data Science |

RESEARCH PROJECT EXPERIENCES

Ski rental with distributional predictions of unknown quality

- Paper accepted at International Conference on Machine Learning (ICML 2026)
- Designed robust learning-augmented algorithms to the classical ski rental problem with any distributional prediction
- Proved high-level upper bound and graceful degradation of algorithm performance with unknown increasing error
- Developed consistency-robustness guarantees, Pareto-optimality results and tight lower bounds.

Controlling tail risk in two-slope ski rental

- Paper accepted at Approximation and Online Algorithms (WAOA 2025) and invited to special issue of Acta Informatica
- Developed technical theorems to fully understand the structure of optimal solution(s) for the problem
- Designed two algorithms to compute the optimal solution(s) for the problem

β -expansions

- Finished a long report about theorems and properties of β -expansions from dynamical systems viewpoint
- Exploited the number of G -expansions, established a sophisticated theorem and gave a five page formal proof
- Studied Uniqueness expansions of points, provided new proofs of lemmas and theorems

PROFESSIONAL SKILLS

- Programming Languages: Python (Pandas, Numpy), Java
- Office Tools: Microsoft Office (Excel, Word, PowerPoint), Google Workspace (Docs, Sheets, Slides)
- Math Tools: LaTeX, MATLAB
- Currently learning: SQL, Tableau, C++

HONORS AND AWARDS

- Bronze medal in 28th International Mathematics Competition for University Students
- The Advance of Algorithm Contest in the Second Round on the Rank List of Facebook Hacker Cup
- 2019-2020, 2020-2021 University of Liverpool Excellence Scholarship (1/320)
- 2019 Summer Undergraduate Research Fellowship

EXTRACURRICULAR / ACADEMIC SERVICE

- Co-instructor of the course EN.553.803 Financial Computing Workshop (Intro to Agentic AI) at JHU.
- Research Assistant in Algorithms and Complexity group at the Johns Hopkins University from August 2023
- Teaching Assistant in Numerical linear Algebra, Approximation Algorithms from 2022
- University of Oxford Mathematics Admission Test (MAT) Grader

HOBBIES AND INTERESTS

- Playing the piano, listening to music, reading
- Travelling the world, hiking, running, playing ball games